

EH Project - Debt Recovery Analysis

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This project has 2 core objectives:

1. Risk Segmentation In insurance, the goal is to group people by risk. In this project, we are analysing hospital claims to see which demographics are most likely to result in an invoice being written off or a late payment of it.
2. Process Optimisation By identifying which demographics that prove to be high relative risk, we can adjust financial reserves or pricing models, or perhaps more realistically for the Revenue Services department at Eastern Health, new processes and methods could be developed to best combat these invoices not being paid.

Additionally, as we will be using a simulated dataset, observed relationships (e.g. age and write-off rates) are driven by underlying modelling assumptions rather than empirical estimation. Therefore, insights should be interpreted as demonstrations of modelling capability rather than real-world findings. Data has been mutated to create a story to analyse.

```
# Creating data we can use for the project
set.seed(2026)
n_records <- 1000

hospital_data <- tibble(
  Invoice_ID = paste0("INV-", 1:n_records), #paste0 attaches the "INV-" to each number (e.g., "INV-1")
  #1:n_records makes it so that the invoices will be numbered 1-1000
  Location = sample(c("Box Hill", "Maroondah", "Angliss", "Wantirna", "Healesville"), #selected major h
    n_records, replace = TRUE, prob = c(0.4, 0.25, 0.15, 0.15, 0.05)), #ensures 40% of
  Patient_Age = sample(1:99, n_records, replace = TRUE), #randomly selects an age between 1 and 99 for
  Claim_Amount = round(rlnorm(n_records, meanlog = 4, sdlog = 0.95), 2), #chosen to use a Log-Normal di
  Date_Billed = sample(seq(as.Date('2025-10-01'), as.Date('2026-04-01'), by="day"), #spreads the invoic
    n_records, replace = TRUE) #the sample function randomly picks one of those date
) %>%
mutate(
  # --- ASSUMPTION 1: Location & Age affect Status ---
  Status = case_when(
    # 1. RURAL PENDING BOOST: Healesville/Angliss get 50% Pending rate
    Location %in% c("Healesville", "Angliss") & runif(n()) < 0.15 ~ "Pending",

    # 2. Written Off (Age-based)
    Patient_Age > 75 & runif(n()) < 0.15 ~ "Written Off",
    runif(n()) < 0.04 ~ "Written Off",

    # 3. Pending (Age-based/Baseline for everyone else)
```

```

Patient_Age > 75 & runif(n()) < 0.35 ~ "Pending",
runif(n()) < 0.10 ~ "Pending",

# 4. Majority is Paid
TRUE ~ "Paid"
),

# --- ASSUMPTION 2: Location affects Payment Speed ---
Base_Delay = sample(0:30, n(), replace = TRUE),

# The "Location Adjustment" mutation for timing
Location_Adj = if_else(Location %in% c("Healesville", "Angliss"),
                        sample(5:15, n(), replace = TRUE), 0),

Days_to_Pay = Base_Delay + Location_Adj
)

head(hospital_data, 3)

## # A tibble: 3 x 9
##   Invoice_ID Location Patient_Age Claim_Amount Date_Billed Status Base_Delay
##   <chr>      <chr>      <int>      <dbl> <date>      <chr>      <int>
## 1 INV-1      Angliss          67         41.2 2026-01-24 Paid         22
## 2 INV-2      Maroondah        75         21.7 2026-02-26 Pending        18
## 3 INV-3      Box Hill         36        135. 2026-01-06 Paid         25
## # i 2 more variables: Location_Adj <dbl>, Days_to_Pay <dbl>

tail(hospital_data, 3)

## # A tibble: 3 x 9
##   Invoice_ID Location Patient_Age Claim_Amount Date_Billed Status Base_Delay
##   <chr>      <chr>      <int>      <dbl> <date>      <chr>      <int>
## 1 INV-998    Angliss          96         18.2 2026-01-24 Pending         9
## 2 INV-999    Box Hill         85         67.3 2025-12-30 Paid            3
## 3 INV-1000   Box Hill         51        259. 2025-12-13 Pending        27
## # i 2 more variables: Location_Adj <dbl>, Days_to_Pay <dbl>

# Plot 1: Distribution of Claims
distribution_claims <- ggplot(hospital_data, aes(x = Claim_Amount)) +
  geom_histogram(binwidth = 10, fill = "steelblue", color = "white") +
  coord_cartesian(xlim = c(0, 500)) + # Zooming in to $500 so you can actually see the peak
  labs(title = "Distribution of Claim Amounts",
       subtitle = "Demonstrating the Log-Normal 'Long Tail' for Hospital Billing",
       x = "Claim Amount ($)",
       y = "Number of Invoices") +
  theme_minimal()

# Plot 2: Patient Age Distribution
plot_age_hist <- ggplot(hospital_data, aes(x = Patient_Age)) +
  geom_histogram(binwidth = 5, fill = "steelblue", color = "white") +
  labs(title = "Patient Age Distribution", x = "Age", y = "Amount of Patients") +
  theme_minimal()

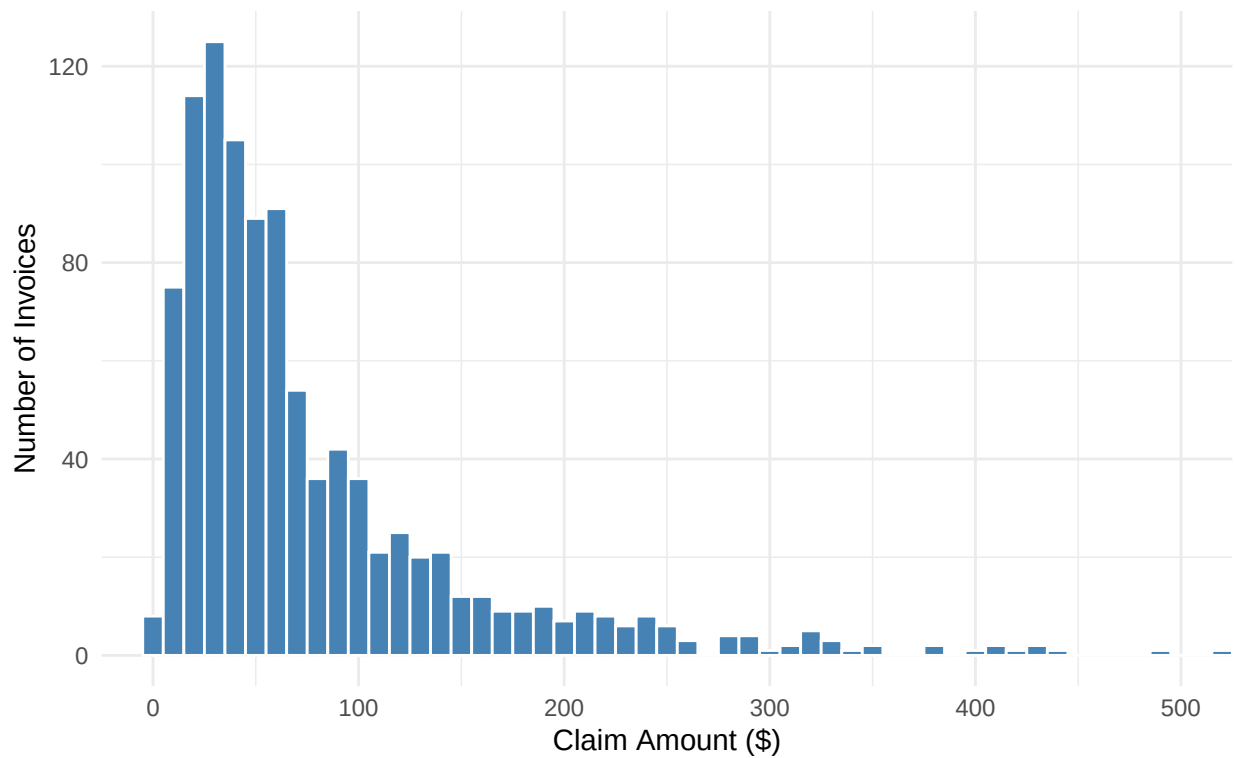
```

```
# Plot 3. Total invoices at each hospital
total_invoices <- ggplot(hospital_data, aes(x = Location, fill = Location)) +
  geom_bar() +
  labs(
    title = "Total Invoice Volume by Hospital Location",
    subtitle = "Comparing total claims across the 1,000 record dataset",
    x = "Hospital",
    y = "Number of Invoices"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

distribution_claims

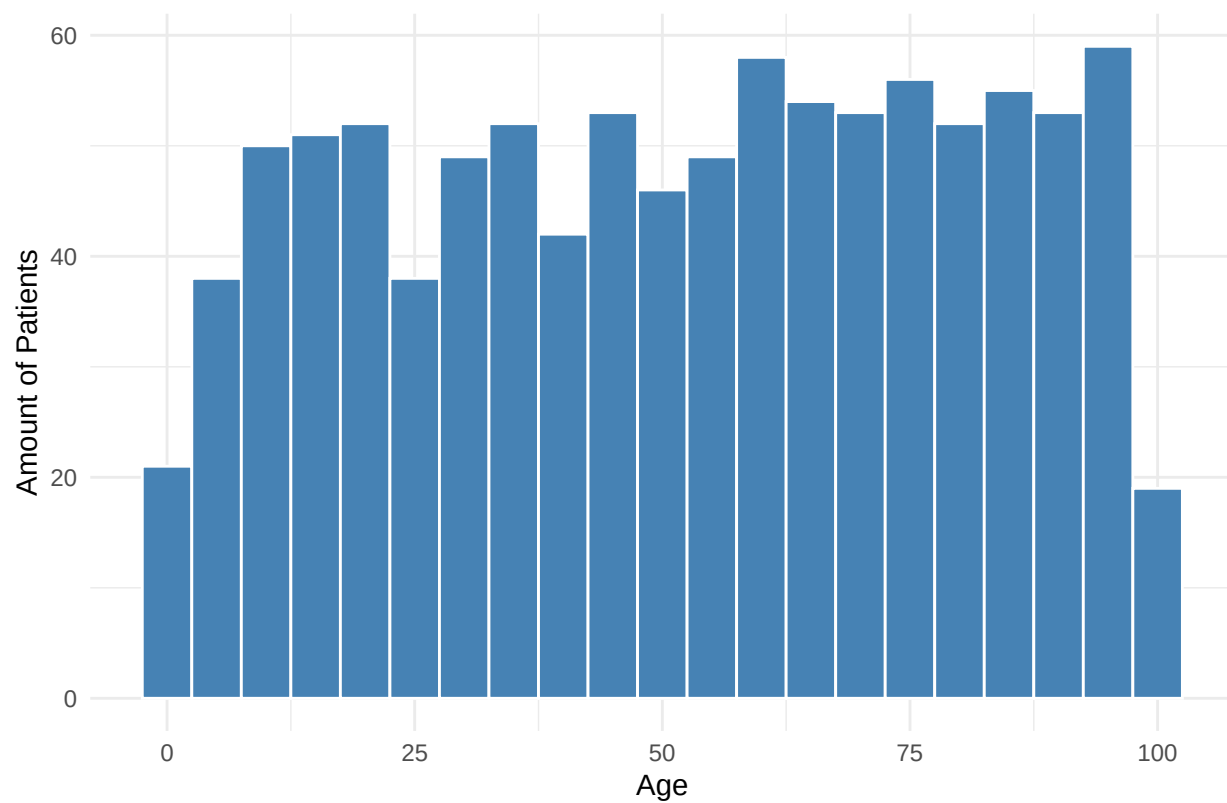
Distribution of Claim Amounts

Demonstrating the Log-Normal 'Long Tail' for Hospital Billing



plot_age_hist

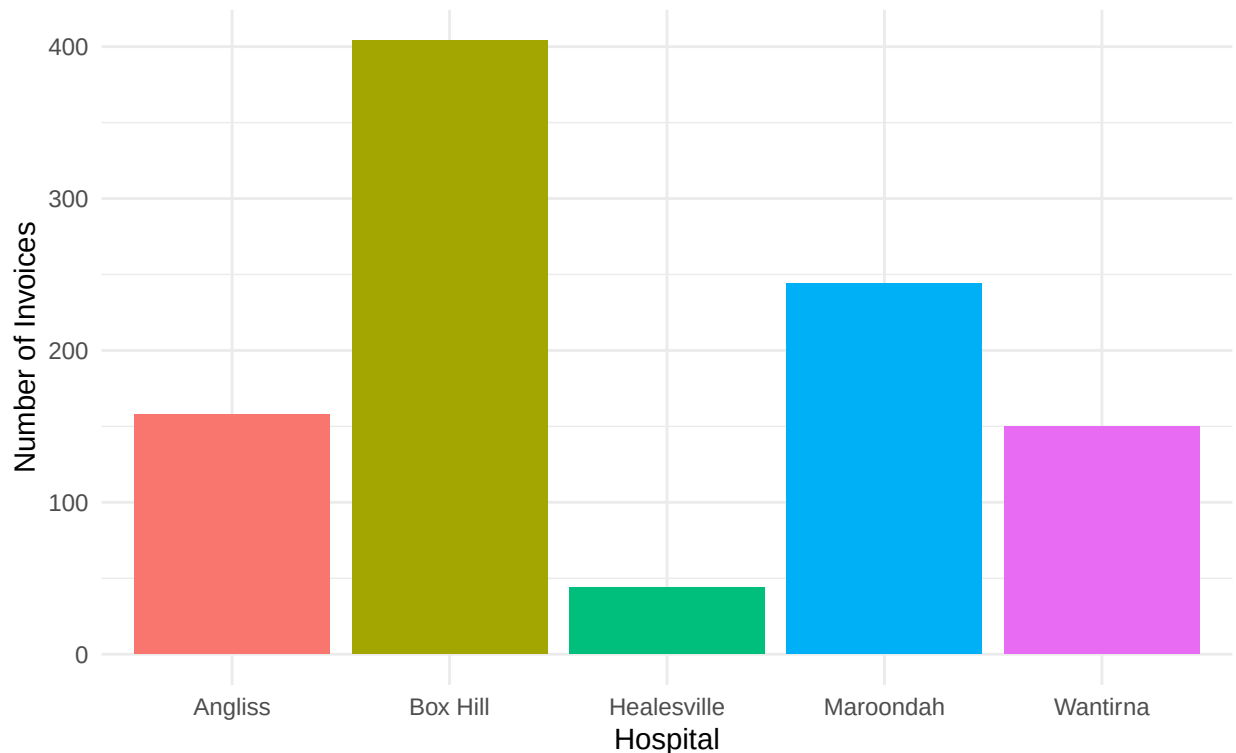
Patient Age Distribution



total_invoices

Total Invoice Volume by Hospital Location

Comparing total claims across the 1,000 record dataset



Looking at the first plot the Distribution of Claims, we can see that the majority of claims are concentrated in the low-value range, with a significant peak occurring between 20 dollars and 60 dollars. The distribution is heavily positively skewed, with the frequency dropping sharply after 60 dollars, the tail extends significantly toward the 500 dollar mark.

Looking at the Patient Age Distribution, we can see a fairly even distribution of patient age, maybe with a slight upward trend.

Looking at the third plot the Total invoices at each hospital, we can see a large amount of invoices being sent out from Box Hill at approximately 400 of the 1000 invoices we've generated. As the hospital get further and further from the city in more rural areas, we can see the amount of invoices decrease.

Next lets see how many of these invoices are actually paid.

```
# 1. Calculate the total counts for each status
status_counts <- hospital_data %>%
  count(Status) %>%
  mutate(percentage = n / sum(n) * 100)

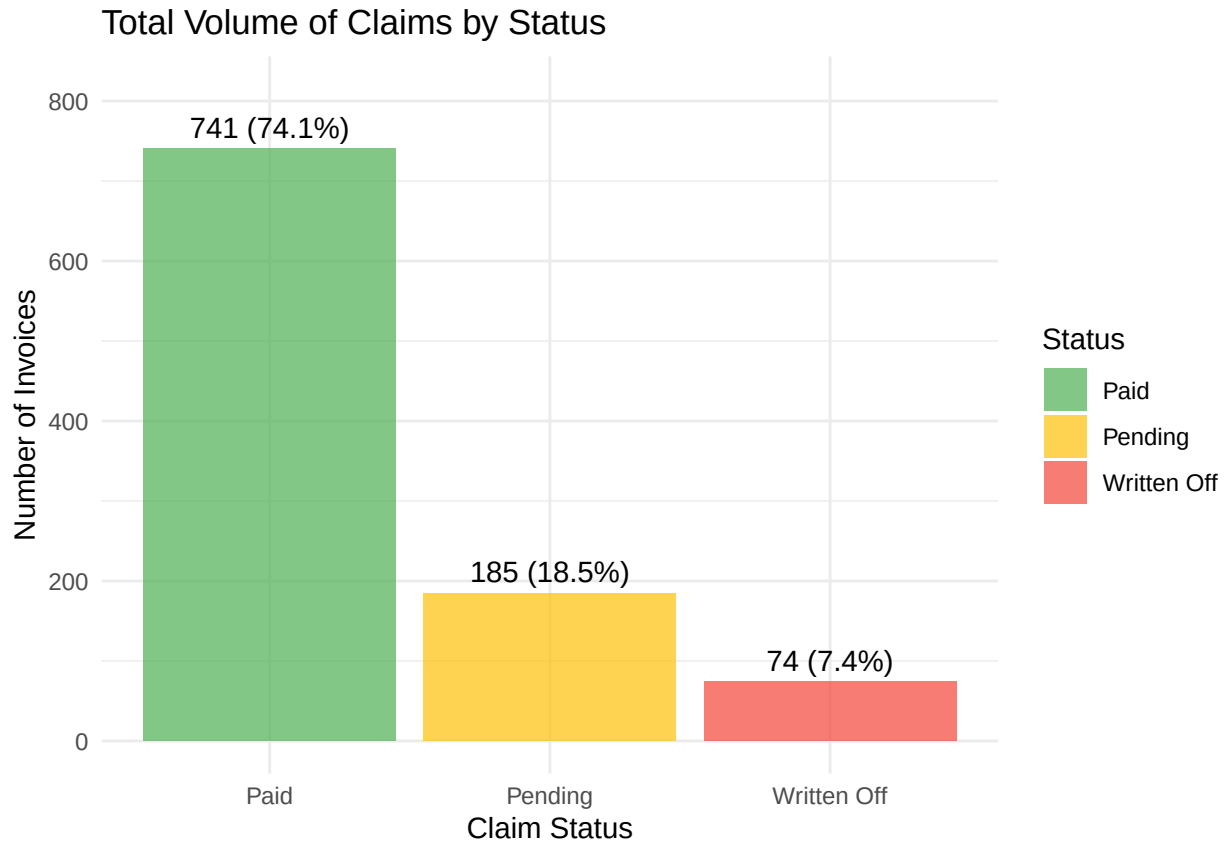
# 2. Create the Bar Chart
plot_status_volume <- ggplot(status_counts, aes(x = Status, y = n, fill = Status)) +
  geom_col(alpha = 0.7) +
  geom_text(aes(label = paste0(n, " (", round(percentage, 1), "%)")),
    vjust = -0.5) +
  scale_fill_manual(values = c("Paid" = "#4CAF50",
                                "Pending" = "#FFC107",
                                "Written Off" = "#F44336")) +
  labs(title = "Total Volume of Claims by Status",
```

```

x = "Claim Status",
y = "Number of Invoices") +
theme_minimal() +
ylim(0, max(status_counts$n) * 1.1) # Add space for labels

```

plot_status_volume



We can see that out of 1,000 records, 741 are successfully paid, 185 pending and 74 written off. Our goal is to analyse whether certain demographics a big influence in the invoices being pending or written.

Next lets see whether age affects whether or not the invoice will be paid.

```

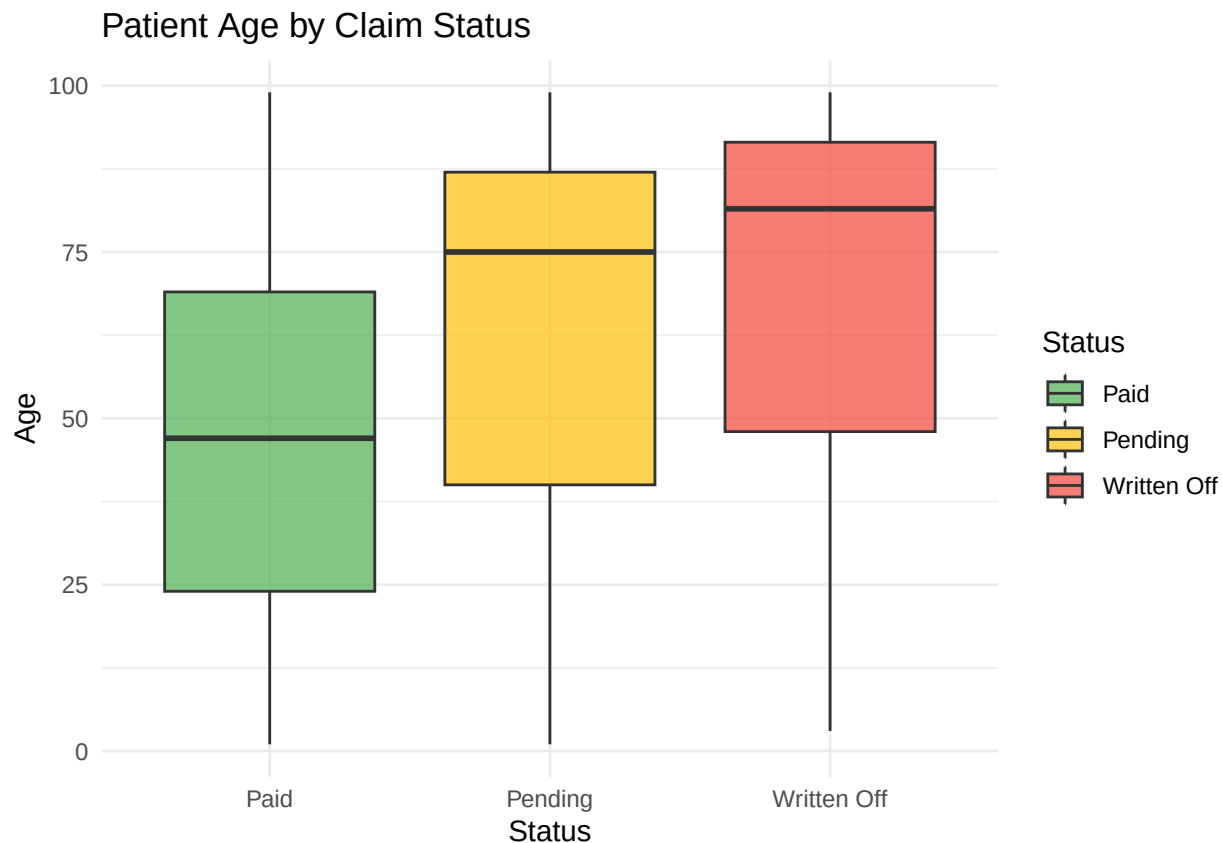
# Plot 1: Boxplot - Age vs Status
plot_age_status <- ggplot(hospital_data, aes(x = Status, y = Patient_Age, fill = Status)) +
  geom_boxplot(alpha = 0.7) +

  # Assigning the colors: Paid = Green, Pending = Yellow, Written Off = Red
  scale_fill_manual(values = c("Paid" = "#4CAF50",
                              "Pending" = "#FFC107",
                              "Written Off" = "#F44336")) +

  labs(title = "Patient Age by Claim Status", y = "Age", x = "Status") +
  theme_minimal()

plot_age_status

```



We can see that the paid box plot is evenly distributed, with a median age of approximately 48 and an interquartile range of 26 to 70.

The pending category shows an upward shift in age with a median age of approximately 60 and an IQR sitting between roughly 30 and 80.

We see an even bigger jump in age with the written off category shows a clear upward shift in age. The median age for written-off claims is approximately 80, with an IQR sitting between roughly 42 and 89.

Older patient demographics appear to be a substantial cause of revenue loss. This suggests that billing for older patients is inherently more complex, potentially due to a problem with delivering the invoice. Modern day billing involves a heavy reliance on emailing and texting. Perhaps if the patient is above a certain age posting could be an automatic method of delivery.

Plot 2: Boxplot - Location vs Status

```
hospital_data_interactive <- hospital_data %>%
  mutate(Status = fct_relevel(Status, "Written Off", "Pending", "Paid")) %>% #reordering so it makes more sense
  group_by(Location, Status) %>%
  summarise(count = n(), .groups = "drop") %>% #we count how many rows in each group and then to forget the groups
  group_by(Location) %>% #We regroup the data, but only by Location this time to see the total for that location
  mutate(percentage = count / sum(count) * 100) #calculating percentage

# creating the ggplot object with custom text for later
p <- ggplot(hospital_data_interactive, aes(x = Location, y = count, fill = Status,
                                           text = paste0("<br>Count: ", count,
                                                         "<br>Percentage: ", round(percentage, 2), "%")))
geom_col(position = "fill", alpha = 0.9) +
```

```

scale_y_continuous(labels = scales::percent) +

# 3. Apply the original "Traffic Light" Color Palette
scale_fill_manual(values = c("Paid" = "#4CAF50",
                             "Pending" = "#FFC107",
                             "Written Off" = "#F44336")) +

labs(title = "Interactive Revenue Risk Profile by Suburb",
     y = "Proportion of Total Claims",
     x = "Location") +
theme_minimal()

# If in RStudio OR knitting to HTML, use hover, otherwise just show the static plot so it knits properly
if (interactive() || knitr::is_html_output()) {
  plotly::ggplotly(p, tooltip = "text")
}

```

The distribution of claim statuses across the locations does not indicate drastic variance in revenue performance between different suburbs, however significant differences can be seen.

Healesville represents a substantial area of concern, featuring the lowest recovery rate at 66% and a disproportionately high pending rate of 20%.

In contrast, Box Hill demonstrates perhaps the strongest financial performance, with 75% of claims successfully paid. However they do have more claims written off than Healesville. This could mean that despite Healesville patients being very slow in paying invoices, they may eventually result in a higher paid proportion than Box Hill.

The high pending rate in Healesville highlights a localised challenge. Quite a few reasons could be the cause. It could be potentially due to slower documentation turnaround or more complex insurer interactions. Additionally, postal time could also be a factor as to why some locations have more invoices pending than others. Since older patients prefer receiving postal invoices, it could take longer for them to pay if the postal takes longer to deliver. Perhaps some hospitals aren't as efficient as others in sending out postal invoices. This will have to be investigated further through process reviews.

Box Hill serves as the benchmark for hospitals; standardising its processes across regional sites like Healesville could help minimise the risk seen in the high pending volume.

Next we'll look into if certain locations are dominated by a certain age group, and whether or not this affects payment.

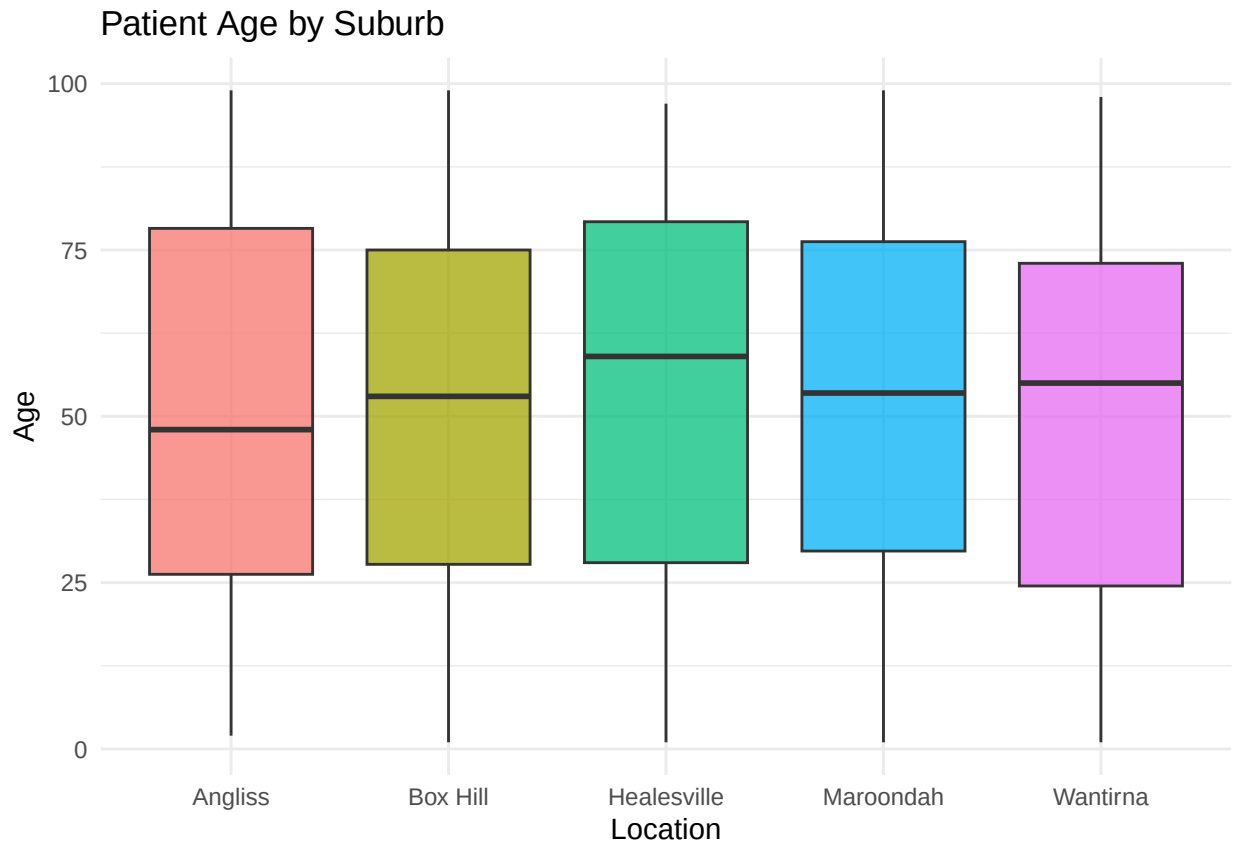
```

# Plot 3: Boxplot - Age vs Location

plot_age_loc <- ggplot(hospital_data, aes(x = Location, y = Patient_Age, fill = Location)) +
  geom_boxplot(alpha = 0.75) +
  labs(title = "Patient Age by Suburb", y = "Age", x = "Location") +
  theme_minimal() + theme(legend.position = "none")

plot_age_loc

```

Healesville stands out with the highest median age at approximately 60 years. The IQR for this location is also shifted higher compared to the other suburbs, spanning roughly from 28 to 80 years. Angliss exhibits the youngest median age at approximately 48 years, while Box Hill, Maroondah, and Wantirna show relatively similar median ages clustered between 53 and 55 years. Across all locations, the age range remains broad, extending from near 0 to nearly 100 years.

Earlier findings established that older patient demographics could be a substantial cause of revenue loss, specifically seeing a clear upward shift in the median age to 80 for claims that have been written off.

The higher median age in Healesville directly correlates with its status as having one of the highest levels of unpaid claims and the most significant in late payments.

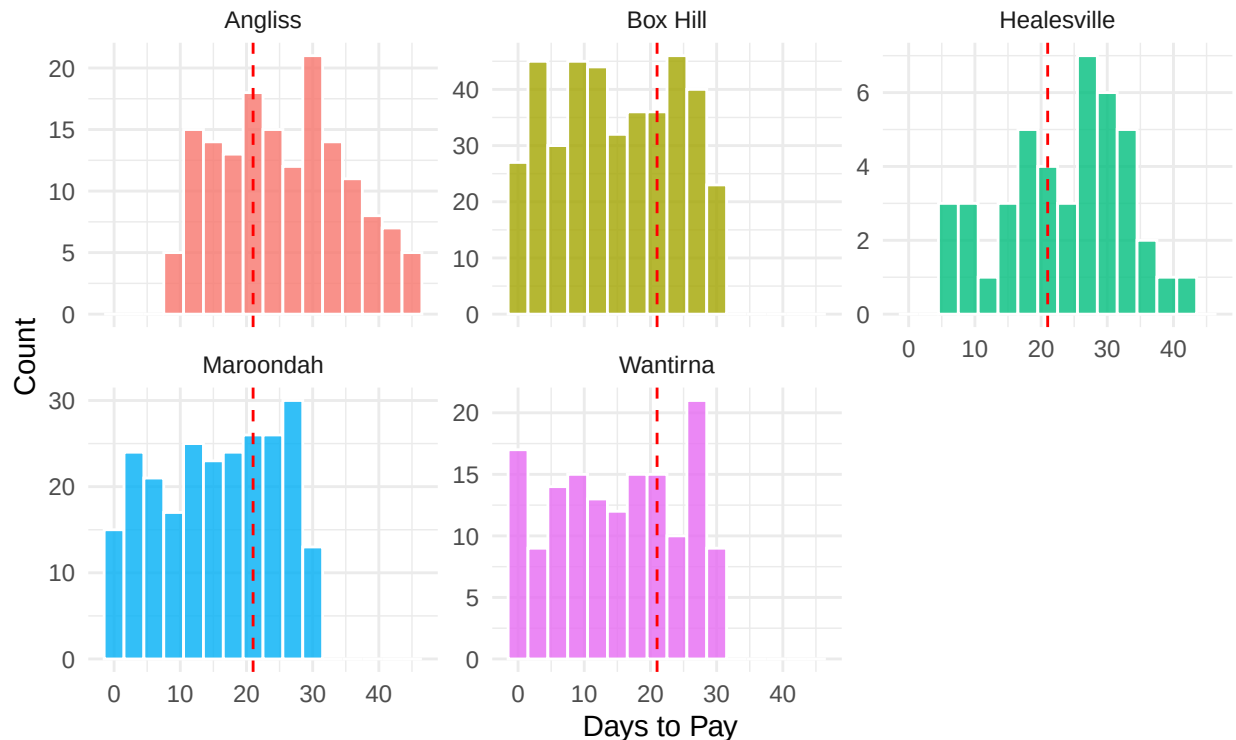
The models suggests that billing for these older demographics is inherently more complex. Since Healesville treats a notably older population, it is disproportionately affected by the “at-risk” status of claims.

```
# Plot 5: Histogram - Payment Speed by Location
plot_loc_hist <- ggplot(hospital_data, aes(x = Days_to_Pay, fill = Location)) +
  geom_histogram(binwidth = 3, color = "white", alpha = 0.8) +
  facet_wrap(~Location, scales = "free_y") +
  geom_vline(xintercept = 21, linetype = "dashed", color = "red") +
  labs(title = "Payment Latency by Suburb", subtitle = "Red line = 21-day Due Date", x = "Days to Pay",
  theme_minimal() + theme(legend.position = "none")

plot_loc_hist
```

Payment Latency by Suburb

Red line = 21-day Due Date



The most significant observation is the rightward shift in distributions for the more rural locations, Healesville and Angliss, compared to the more centrally located facilities. In Healesville, the data reveals a critical issue; despite having the lowest volume of the group, almost every single claim falls to the right of the 21-day red line. This indicates that rurality acts as a near-guarantee of overdue payment, with the bulk of invoices being paid between 25 and 35 days.

Angliss follows a similar trend, though with a much higher volume of patients than Healesville. The plot shows very few claims are settled within the first 10 days with the peak concentration occurring between 20 and 30 days. This suggests that while Angliss is a busier facility, it shares the same structural bottlenecks as Healesville, consistently pushing the majority of its revenue into the “overdue” category.

In contrast, the suburban hospitals like Box Hill, Maroondah and Wantirna show a more balanced financial profile. Box Hill manages to capture a wide spread of payment times, including a significant portion of early revenue (days 0–15) that the rural sites lack entirely. While it still has a large volume of late payments, the distribution is more uniform, suggesting a more diverse patient base or more efficient billing interactions that aren’t entirely tied to a single delay factor.

Maroondah shows a steady buildup of claims as the 21-day deadline approaches, with a peak after the due date that mirrors Box Hill’s. This could be because overdue bill reminders are sent out around here.

Wantirna displays a sharp spike of early payers who settle within the first 5 days, yet also maintains a secondary larger peak of late payers around the 25-day mark, mirroring the previous 2 hospitals.

```
# 1. Focus only on the 'Paid' claims to see the payment distribution
```

```
paid_only <- hospital_data %>%
```

```
  filter(Status == "Paid")
```

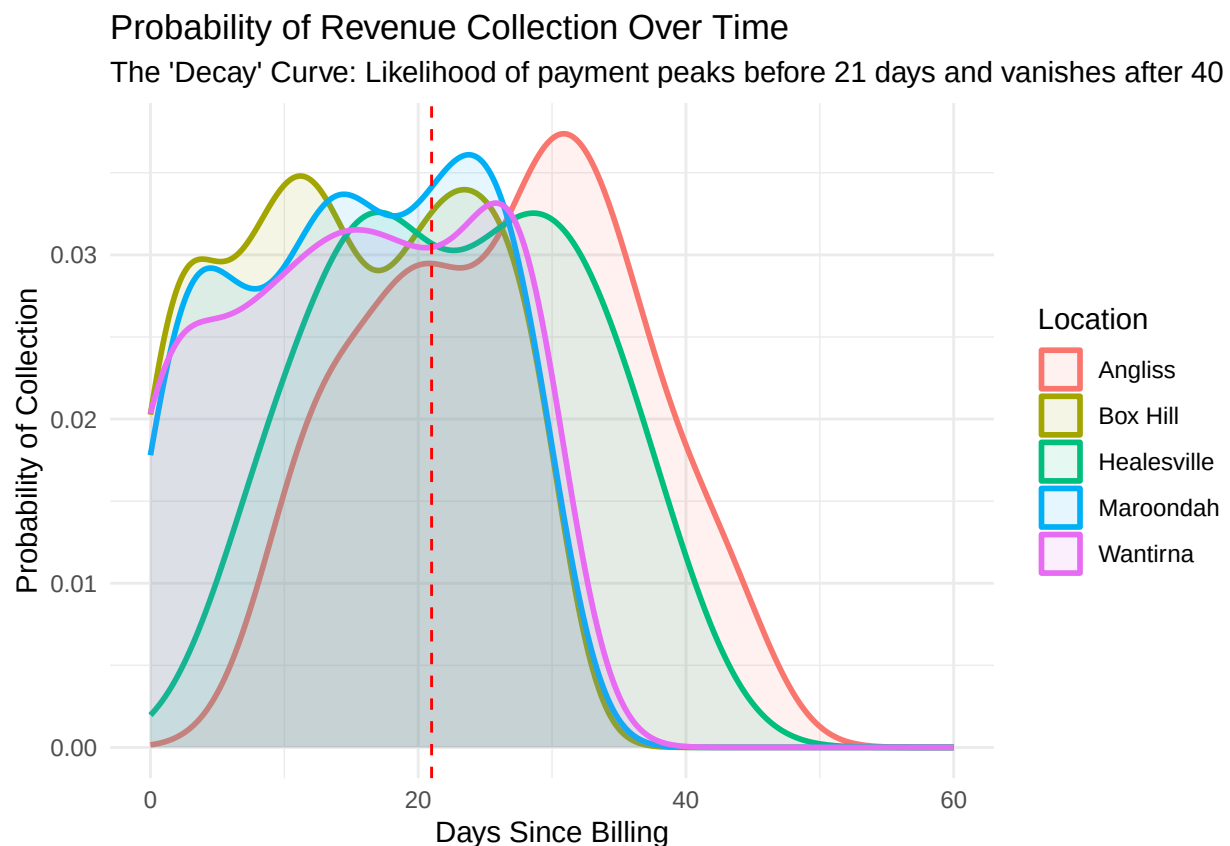
```
# 2. Create a Smooth Density Plot
```

```
# This automatically handles the "unlikely to be paid" decay we're looking for
ggplot(paid_only, aes(x = Days_to_Pay, color = Location, fill = Location)) +
  # 'geom_density' is the key for that smooth "U" or "Bell" shape
  geom_density(alpha = 0.1, size = 1) +

  # Standardising the x-axis to show the 40+ day drop-off
  xlim(0, 60) +
  geom_vline(xintercept = 21, linetype = "dashed", color = "red") +

  labs(title = "Probability of Revenue Collection Over Time",
       subtitle = "The 'Decay' Curve: Likelihood of payment peaks before 21 days and vanishes after 40",
       x = "Days Since Billing",
       y = "Probability of Collection") +
  theme_minimal() +
  theme(legend.position = "right")
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



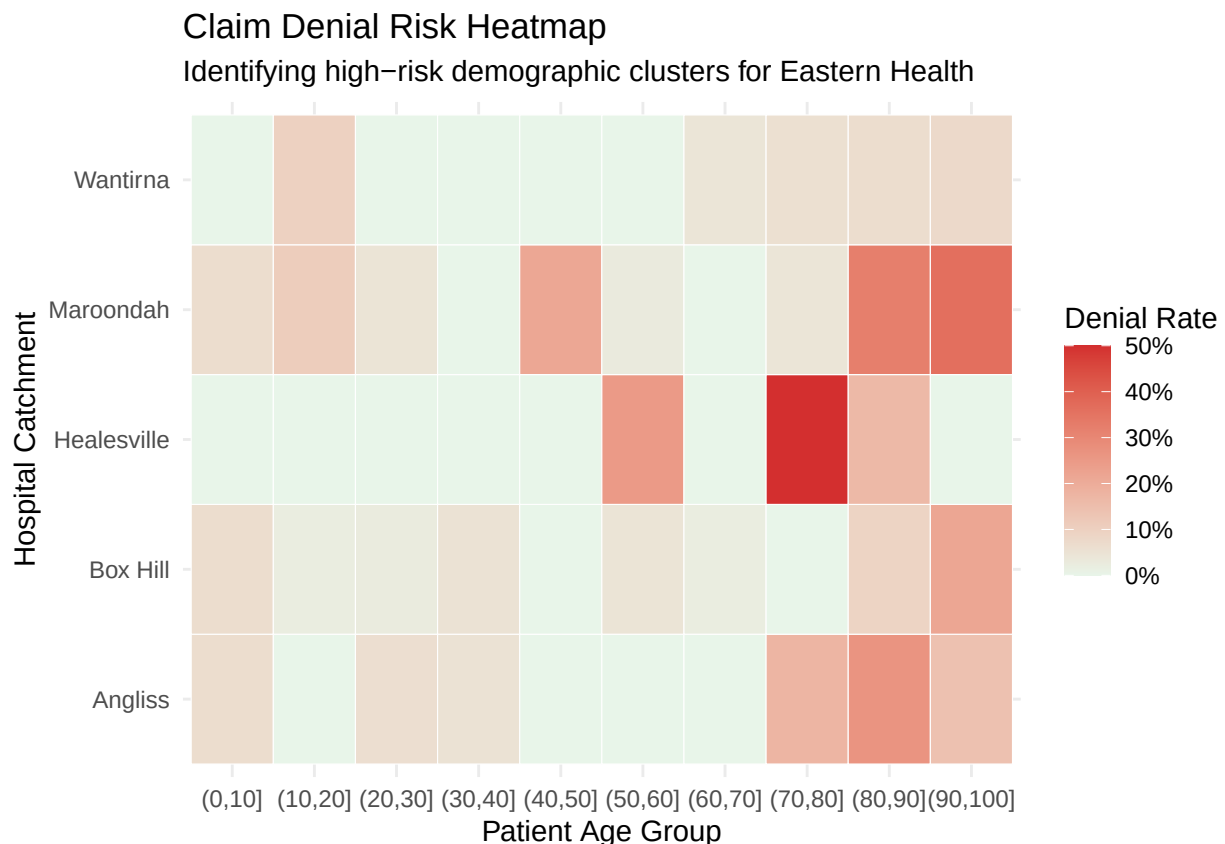
The more suburban locations Box Hill, Maroondah, and Wantirna exhibit high probability densities early in the billing cycle meaning there's a high chance they're paid right after the service. These curves show that a significant portion of their revenue is collected within the first two to three weeks meaning the probability

of collection remains relatively high and stable during the initial period and begins a steep decline once the 30-day mark is reached.

In contrast, Angliss and Healesville display a clear right shift in their probability distributions. Unlike their suburban counterparts, these two locations have very low collection probabilities in the first 10 days. Instead, their likelihood of payment is around and after the 21-day due date. Angliss shows the most extreme delay, with its most likely day its paid is peak occurring between 30 days.

```
# 1. Create Age Brackets
heatmap_data <- hospital_data %>%
  mutate(Age_Group = cut(Patient_Age, breaks = seq(0, 100, by = 10))) %>%
  group_by(Location, Age_Group) %>%
  summarise(Denial_Rate = mean(Status == "Written Off"), .groups = 'drop')

# 2. Plot the Heatmap
ggplot(heatmap_data, aes(x = Age_Group, y = Location, fill = Denial_Rate)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "#e8f5e9", high = "#d32f2f", labels = scales::percent) +
  labs(title = "Claim Denial Risk Heatmap",
       subtitle = "Identifying high-risk demographic clusters for Eastern Health",
       x = "Patient Age Group",
       y = "Hospital Catchment",
       fill = "Denial Rate") +
  theme_minimal()
```



The heatmap identifies specific demographic clusters where revenue loss is most severe. The most dominant trend across the entire Eastern Health network is a sharp increase in denial rates as patients move into the

70-100 age brackets.

The most critical “hotspot” on the map is found in the Healesville for the 70-80 age group. This cell is the darkest red on the entire grid, indicating a denial rate around the 50% mark. This visualises the combination of a rural location and a senior patient demographic creates a perfect storm for claim denials. While other hospitals like Angliss and Maroondah also show high risk in their oldest cohorts (80-100), the intensity in Healesville suggests that being a senior in a rural area is the highest risk factor in the model.

In contrast, the lighter areas of the heatmap covers the majority of the younger demographics, specifically those under the age of 50. For hospitals like Box Hill and Wantirna, the denial risk remains consistently relatively low and stable for all ages of patients, typically hovering near 0-20%. There are minor “blips” in the middle-aged cohorts—such as a slight increase for 40-50 year olds at Maroondah—but these are outliers compared to the overwhelming trend of aging related risk.

Conclusion:

The analysis of the simulated hospital claims reveals that patient age is a significant cause of revenue loss across Eastern Health hospitals. The trends depicted from models showing that older patients tend to have their invoices written off more and more invoices pending could demonstrate that the complexity of modern billing which relies heavily on digital communication like email and text messages create barriers for older demographics. Implementing an automated postal delivery system for patients above a specific age threshold could serve as a practical intervention to ensure invoices are received and understood, potentially reducing the high volume of pending and written-off claims observed in senior cohorts.

Geographic location also plays a critical role in the speed of revenue collection, highlighting a distinct performance gap between suburban and rural facilities. Suburban hospitals like Box Hill function captures a significant portion of early revenue within the first 15 days. In contrast, rural locations such as Healesville and Angliss face structural bottlenecks that push the vast majority of payments past the 21 day due date, with most rural invoices only being settled between 25 and 35 days. This could likely due to slower documentation turnarounds or longer postal delivery times. Internal review will need to be done to understand what is truly occurring.

The most severe area of concern is when senior demographics intersect with rural locations. The data identifies a critical risk hotspot in Healesville specifically for the 70–80 age bracket, where the denial rate peaks at approximately 50% while younger patients under 50 consistently maintain low denial risks regardless of their location, the intensity of revenue loss in older rural populations is disproportionate. To optimise recovery, Eastern Health should focus on adapting to each location and age demographic in order to build a more efficient billing process.